Computer Networks

12th November 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Start your answer to each question on a new page.

Answer ALL questions

Marks

Question 1

Explain the TCP/IP model by answering the following:

a) Describe the primary function of the Internet Layer. **2**

b) Identify a device that operates at the Network Access Layer **and** explain its role. **2**

c) Explain the primary function of Transport Layer in details (including the handshaking procedure) **and** describe how session-related functions such as establishing, maintaining and terminating communications are handled in TCP/IP model. **4**

d) Identify the layer of the TCP/IP model responsible for data encryption and decryption. Provide an example of a protocol that operates at this layer. **2**

Total 10 marks

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Question 2

a) What is the message encapsulation process **and** what is the data called in each OSI layer? **8**

b) Given the following scenario, identify **and** justify the most suitable types of networks to be used (LAN, WAN, MAN) **2**

A global corporation with offices in multiple cities in the UK needs to connect its regional data centres to facilitate secure data transfer and to communicate. The corporation also needs computers, printers and server connections for TWENTY (20) employees within each single office building for file sharing and communication.

Total 10 marks

Question 3

You are managing a small company's LAN with TWENTY (20) computers in a star topology using a single switch. The company plans to expand, adding FORTY (40) more employees and requires wireless connectivity for mobile devices, laptops and visitors. Your task is to redesign the network into a tree topology and integrate wireless access points to accommodate both wired and wireless devices. Ensure the network is scalable, efficient and secure.

a) Explain **THREE (3)** advantages of the tree topology over star topology. **6**

b) Identify the most suitable cabling based on bandwidth needs, distance limits and scalability for high-speed data transfer. Justify your choice. **2**

c) Recommend the best Wi-Fi standard to handle device load, optimise speed, reduce interference and ensure secure access for all users. Justify your choice. **2**

Total 10 marks

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Question 4

a) Explain what a security attack is. **1**

b) For each of the following incidents, identify **and** explain the type of attack. Additionally, recommend **TWO (2)** security countermeasures for each incident to prevent similar threats in the future.

i) **Incident 1**: An attacker intercepted sensitive customer data, including personal and financial details, while it was being transmitted over the network. **3**

ii) **Incident 2:** An organised attack overwhelmed the company's web server, making it unavailable to customers for several hours. **3**

iii) **Incident 3:** An employee inadvertently downloaded an infected email attachment, which subsequently locked the system and demanded a payment for decryption. **3**

Total 10 marks

Question 5

a) This question is about firewall and intrusion detection system (IDS).

i) Define a firewall **and** explain its primary purpose in network security. **2**

ii) Define an Intrusion Detection System (IDS) **and** explain its primary purpose in network security. **2**

iii) Compare **and** contrast the roles of firewalls **and** IDS in network security. **2**

b) List **and** briefly explain **TWO (2)** types of firewalls in regard to their functionality. **4**

Total 10 marks

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Question 6

You are the network administrator for a medium-sized company with approximately 150 employees operating a local area network (LAN). The company requires secure remote access for employees who need to connect to the company network from outside locations, such as their homes or while travelling.

a) Identify, explain, **and** provide **ONE (1)** example of the types of network software required for this network, including the appropriate system software for servers, client machines **and** network devices. **6**

b) Identify the most suitable option and types for the company's LAN remote connection. **1**

c) Explain THREE (3) fundamental functions of the software identified in b). **3**

Total 10 marks

Question 7

List **and** explain **FIVE (5)** issues of VOIP services noting the problem and their cause. **10**

Total 10 marks

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Question 8

You are tasked with planning the IP address allocation for the following network:

Small Office Network: A small business with **25 devices** (computers, printers, and mobile devices) that needs a private local area network (LAN) for file sharing, printing and internet access. This business has 3 separate departments.

a) Specify the **IP v4 address class** that would be most suitable for the Small Office Network. Justify your selection with regards to nodes allocation. **2**

b) Give example IP address range for each network (network segments) **4**

c) Explain why you chose that specific IP class for each network, considering factors such as: **4**

- Number of devices supported
- Future expansion
- Segmentation requirements
- Private vs. public IP addresses

Total 10 marks

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Question 9

You are the network administrator for a medium-sized office building with multiple floors. The building has an open-plan layout on each floor, along with several enclosed meeting rooms. The management has requested the implementation of a Wireless Local Area Network (WLAN) to support approximately 100 devices, including laptops, smartphones, and tablets. However, during your initial site survey, you observe some challenges:

1. You notice that certain areas of the building, especially the far corners of each floor and enclosed rooms, experience weak or no signal coverage.
2. Thick concrete walls that separate rooms and floors.
3. Numerous electronic devices, such as microwaves and cordless phones, that operate on the frequency bands 2.4 GHz and 5 GHz.
4. Nearby wireless networks from neighbouring buildings that may cause overlapping channels and congestion.
5. You notice an issue of neighbouring building employees connecting to the guest user account of the Wi-Fi.

a) Explain how each of the FIVE (5) challenges identified above can impact effective Wi-Fi implementation. **5**

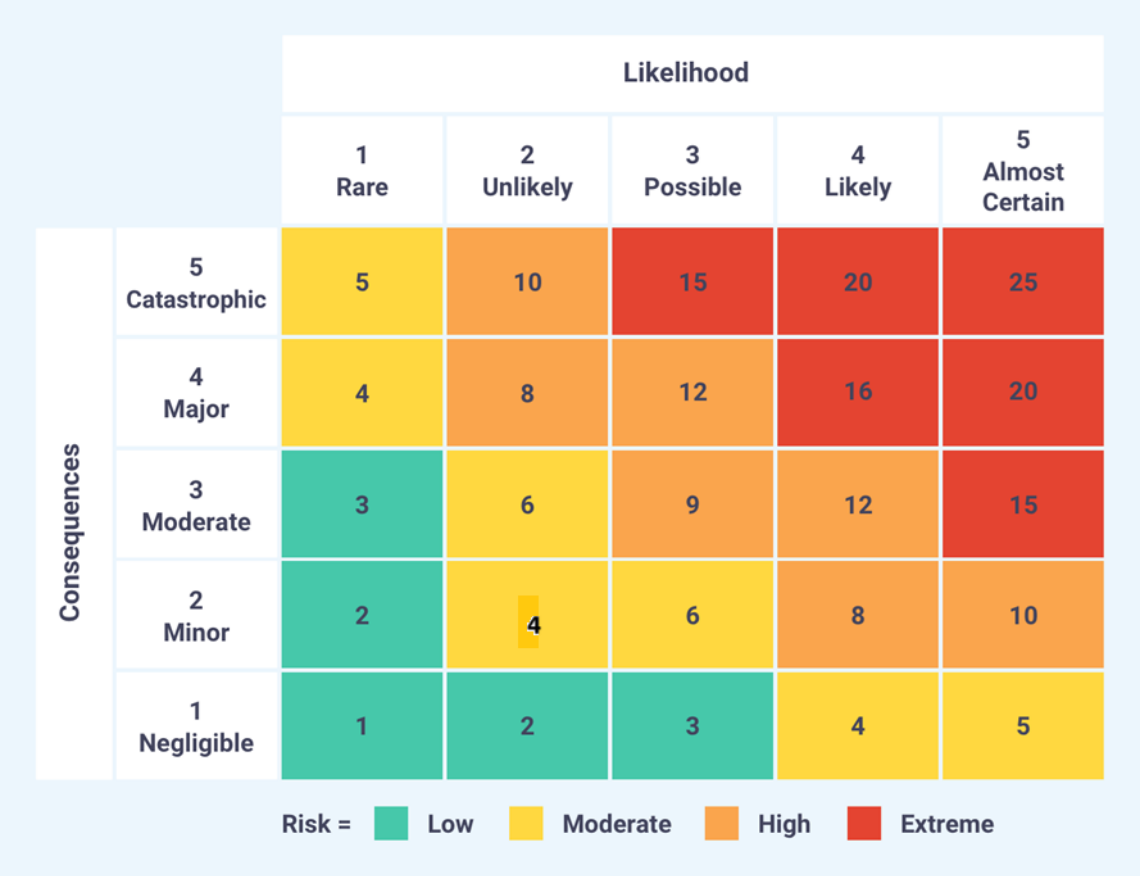
b) Identify **ONE (1)** strategy to resolve each of the challenges you have observed in (a). **5**

Total 10 marks

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Question 10

Below is a Risk Assessment Matrix outlining various risks associated with a company's network infrastructure. **10**



Using the provided Risk Assessment Matrix, try to fill out the following risk assessment spreadsheet by identifying the elements should they be appropriate to each blank cell in the table (2(a), 2(b), etc).

Serial	Asset	Threat	Vulnerability	Loss of CIA	Severity	Likelihood	Risk (1-25)	Action	Control measure
1	Web server	flood	Water damage	A	5	1	5	mitigate	backup
2	Web server	theft	2(a)	2(b)	5	2	2(c)	2(d)	2(e)
3	Web server	Packet sniffing	3(a)	3(b)	4	4	3(c)	3(d)	3(e)

Total 10 marks

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END OF PAPE

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